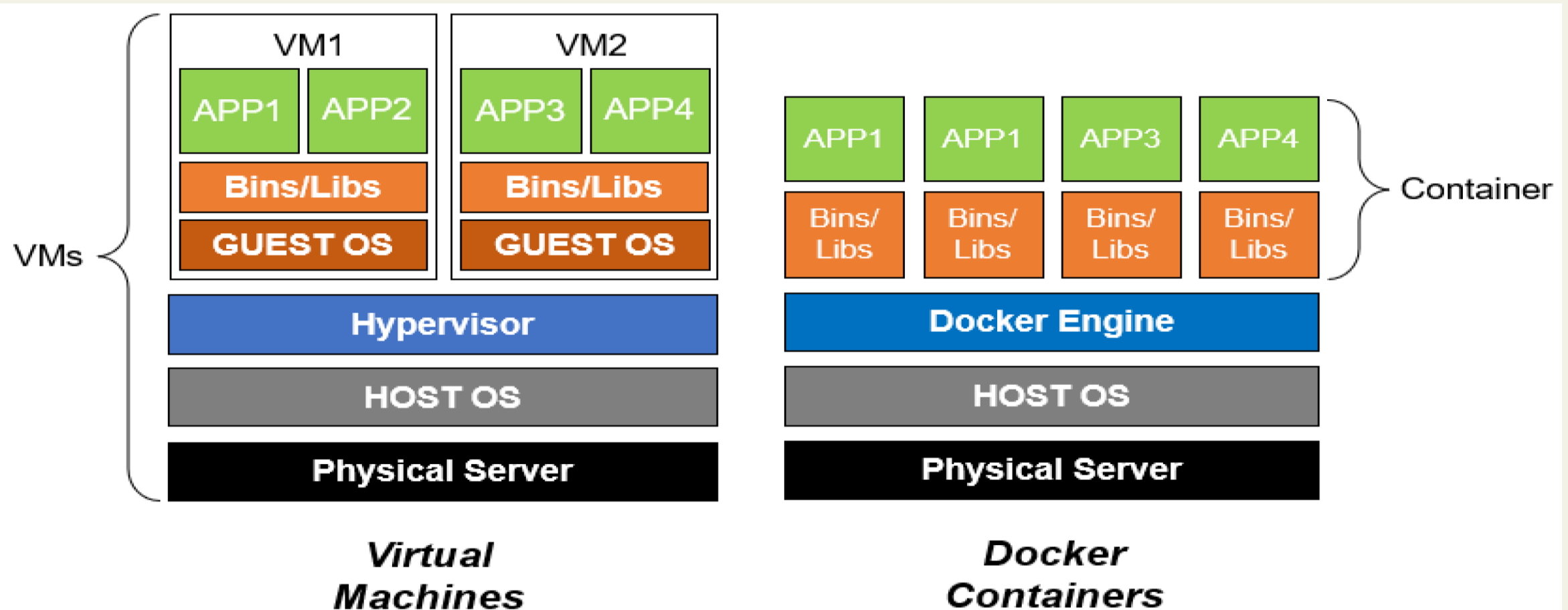

Introduction to Docker

PRESENTED BY: RANA AHMED



Difference between virtual machine and container



Compare:

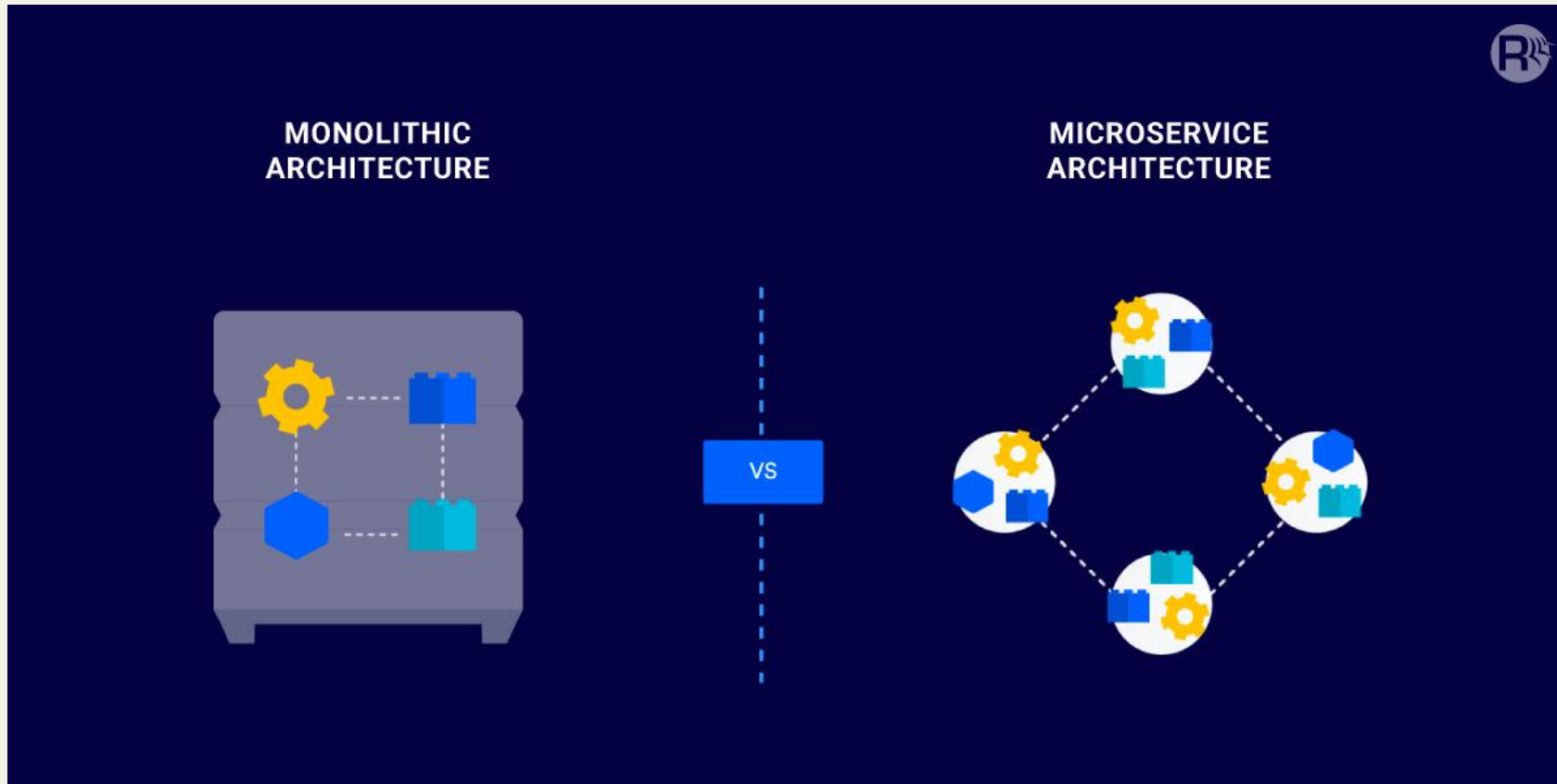
VM

1. Heavy weight
2. Startup time in minutes
3. Each vm runs its own os
4. Vm provider :
 - vmware vsphere
 - Microsoft hyper -v
 - oracle vm virtualbox

CONTAINER

1. Light weight
2. Start in milli second
3. You can scale container in and out
4. Container share the host kernel
5. Container provide:
 - docker
 - Rocket
 - cRI-o
 - containerd

What is the difference between monolith app and microservice :



Monolith and microservice :

Monolith: the entire application is built as a single code project.

- Harder to scale and troubleshoot

Microservices: application code processes or microservices are built as separate code projects.

- Easier to scale and troubleshoot.
- Each backend service exposes a set of APIs through an API Gateway.

(cont.)



Monolith



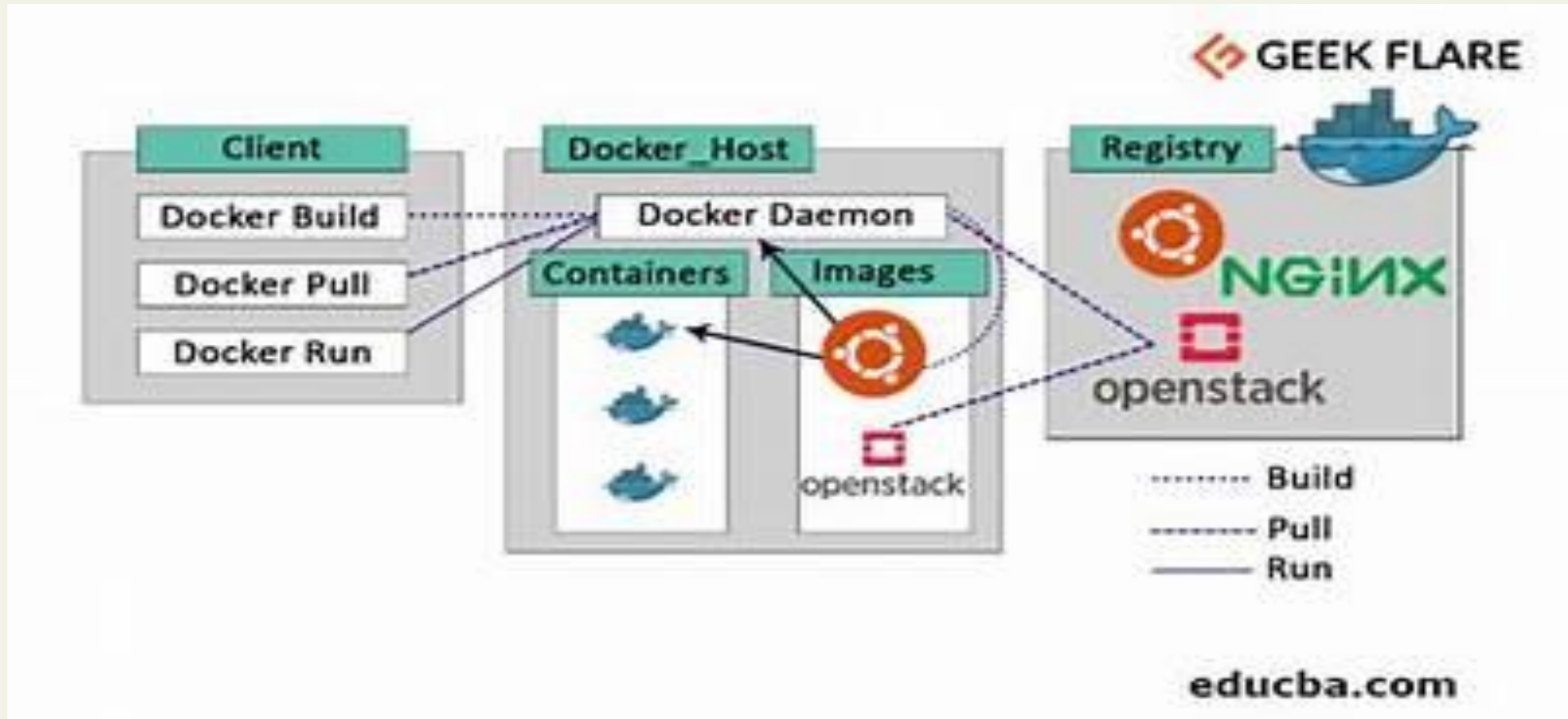
Microservices

What is docker ?

- Docker is a software platform that simplifies the process of building, running, managing, and distributing applications using containers.
- Developed by Docker Inc.
- Docker is an open source.
- Docker virtualizes the host Os (container uses the same kernel).
- Docker allows for running applications in containers (isolated environments).

There are two types of Docker Editions: docker-CE, docker EE

Docker architecture:



Docker Architecture(cont.)

1- `docker cli` :

It enables developers/ users to interact with docker .

2- `docker host` :

A- the daemon (dockerd)

B- images (template)

C- container (encapsulated env)

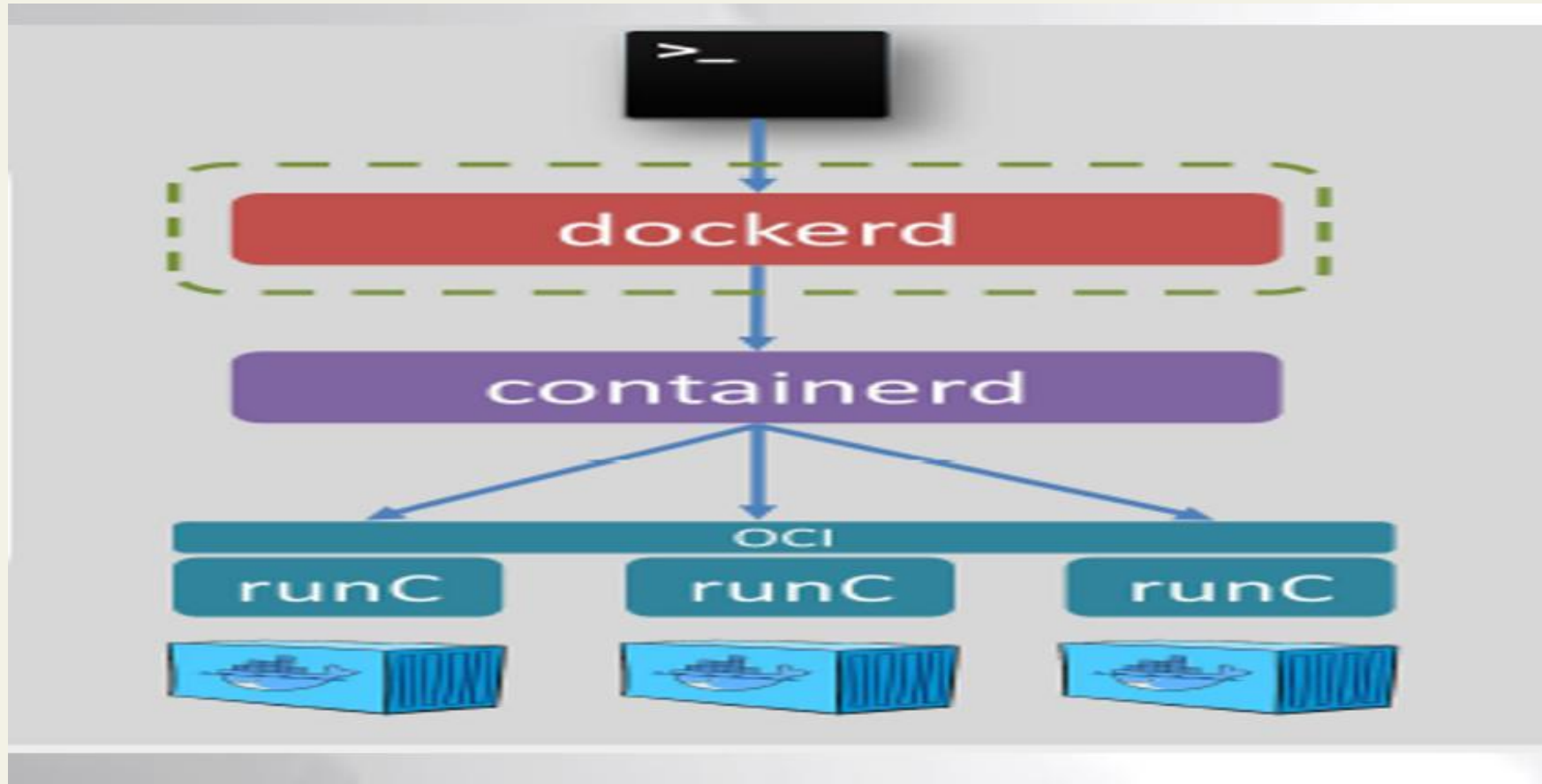
3- `docker registries`

Ex: docker hub

Ex: Amazon elastic container registry (ECR)

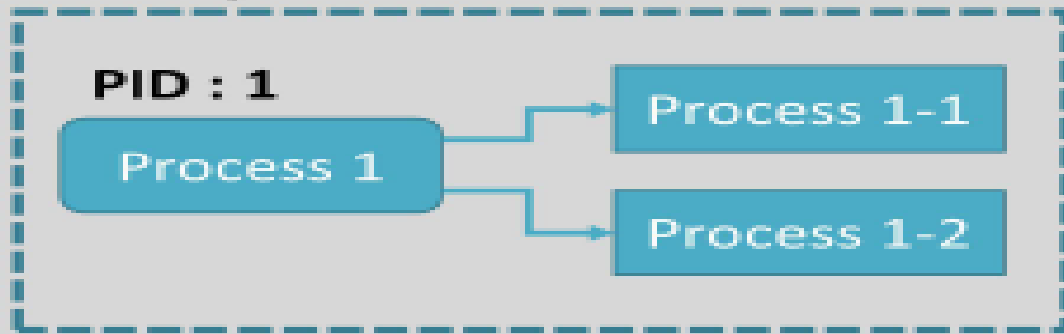
Ex: Azure container registry (ACR)

Understand how to run containers :

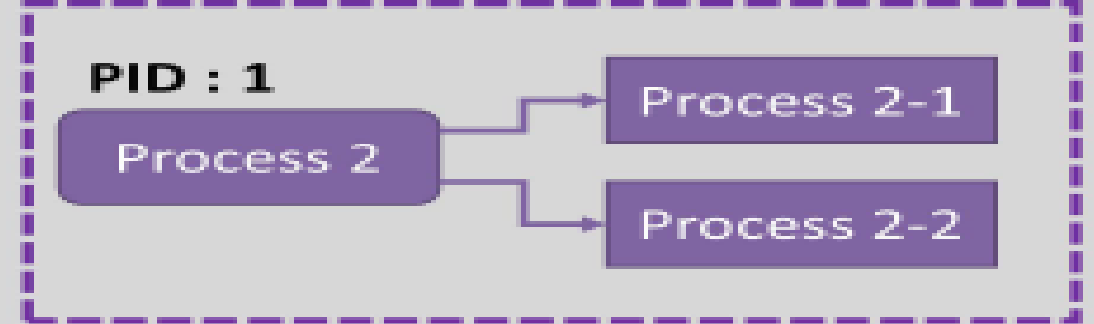


Namespace & c group :

Namespace



Namespace



Cgroups



Installing docker on machine

Go to url : <https://docs.docker.com/engine/install/centos/>

Important : `sudo usermod -aG docker $username`

<<must reboot machine>>

Basic docker commands-part1 :

```
docker pull <image_name >
```

```
Docker create <image-name>
```

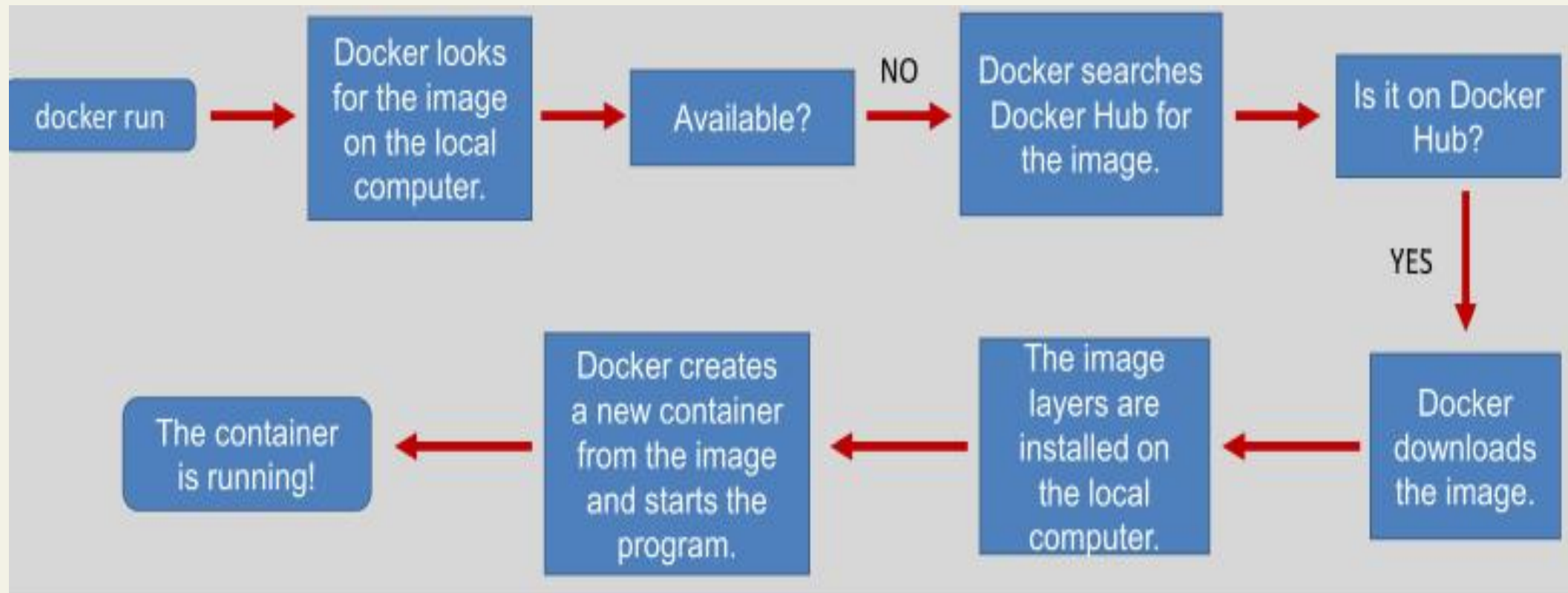
```
Docker start <container-name>
```

```
Docker stop <container-name>
```

Lab1:



Docker run <image_name>



Basic docker commands-part2:

```
docker run -it alpine /bin/bash
```

```
docker run -d centos
```

```
docker run -d centos sleep 20
```

```
docker exec -it <container_id> bash
```

Basic docker commands-part3:

```
docker stats <container_id>
```

```
docker rm <container-name> / docker rmi <image-name>
```

```
docker ps / docker container ls / docker ps -a
```

```
docker images / docker image ls
```

Lab2:



Basic docker commands-part4:

```
docker run -d --name <container-name> <image-name>
```

```
docker inspect <container-id> / docker inspect <image-name>
```

```
docker logs <container-id>
```

```
docker commit <container-id> < new image name>
```

Lab3:



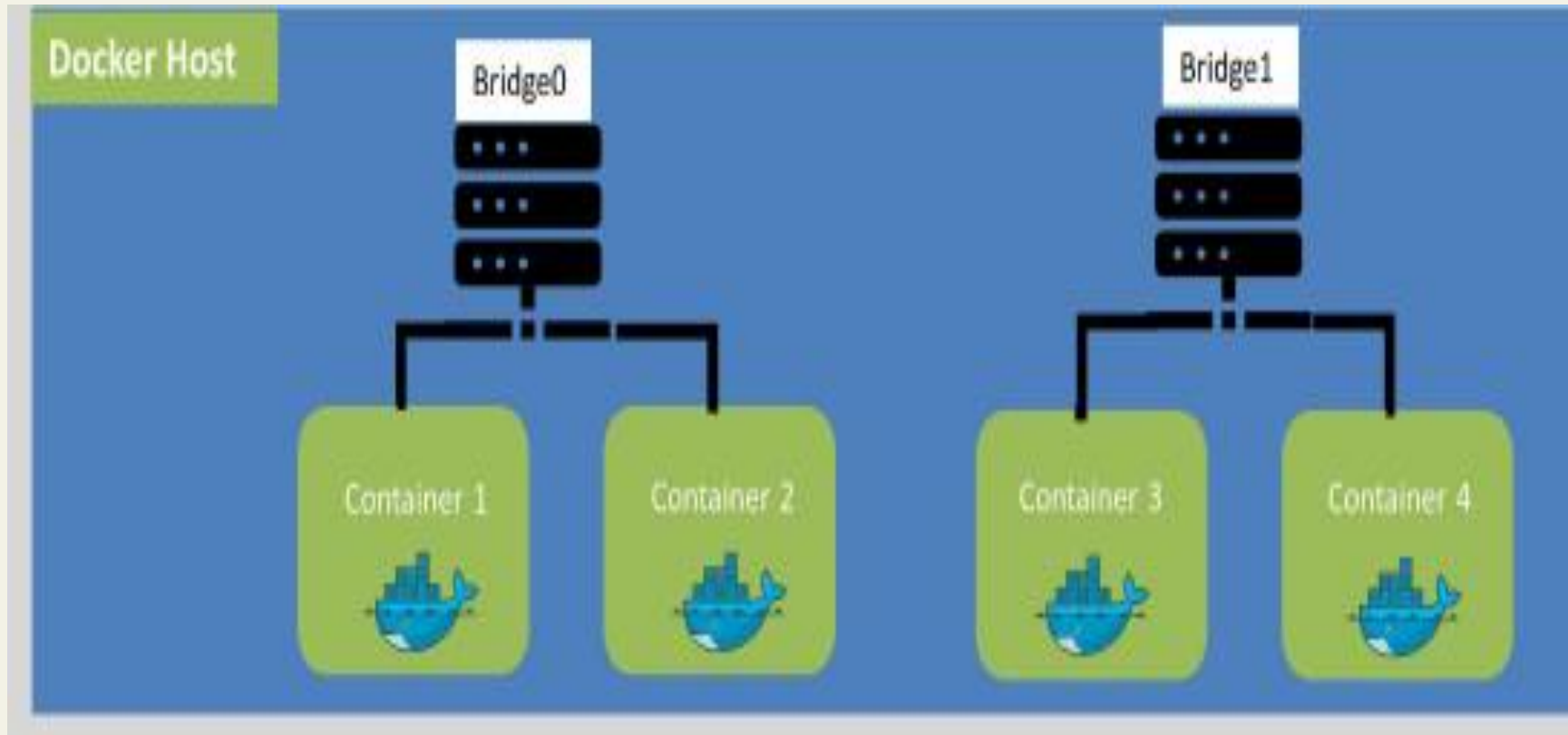
❖ Docker networking-intro:

Docker networking is primarily used to establish communication between docker containers and the outside world via the host machine where the docker daemon is running.

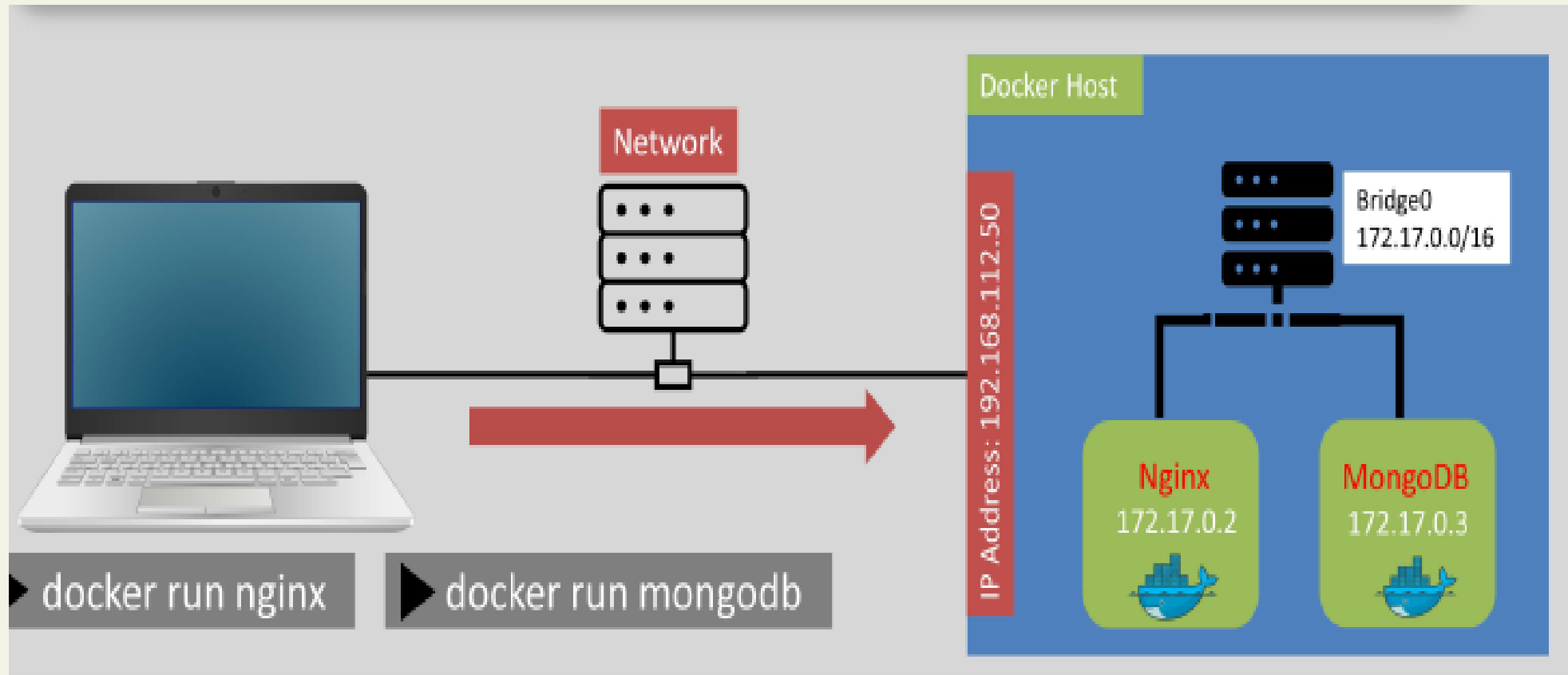
Types of docker network :

- 1- bridge
- 2- host
- 3- none

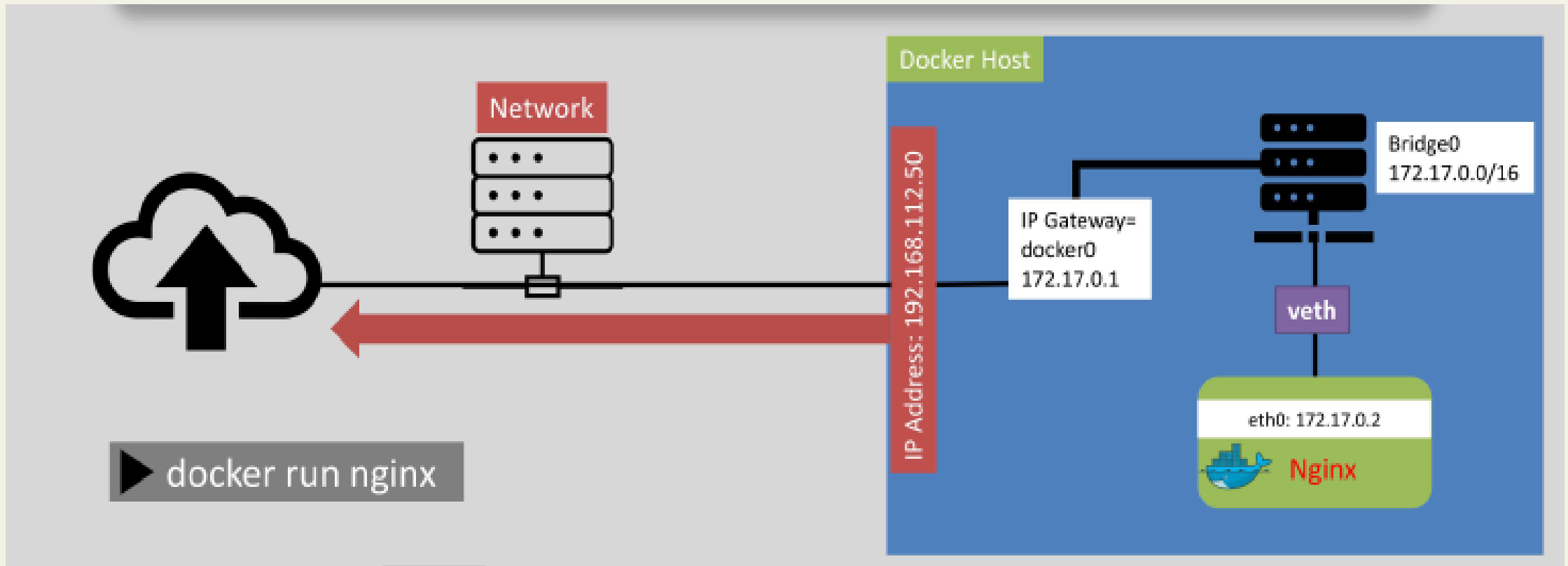
1-Docker networking-bridge network :



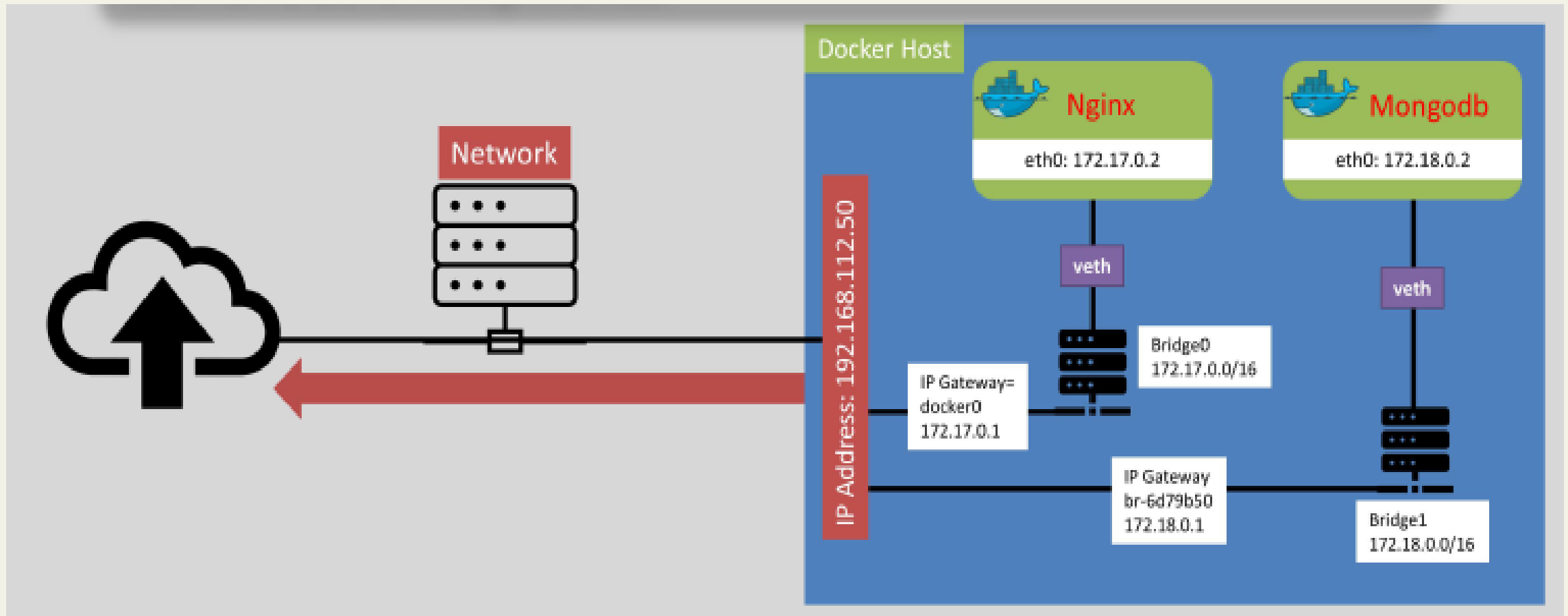
- The docker default ip range for the default bridge network (bridge0) is 172.17.0.0/16
 - 172.17.0.1 is the default gateway between bridge0 and the docker host
-



How containers connect to the External network?

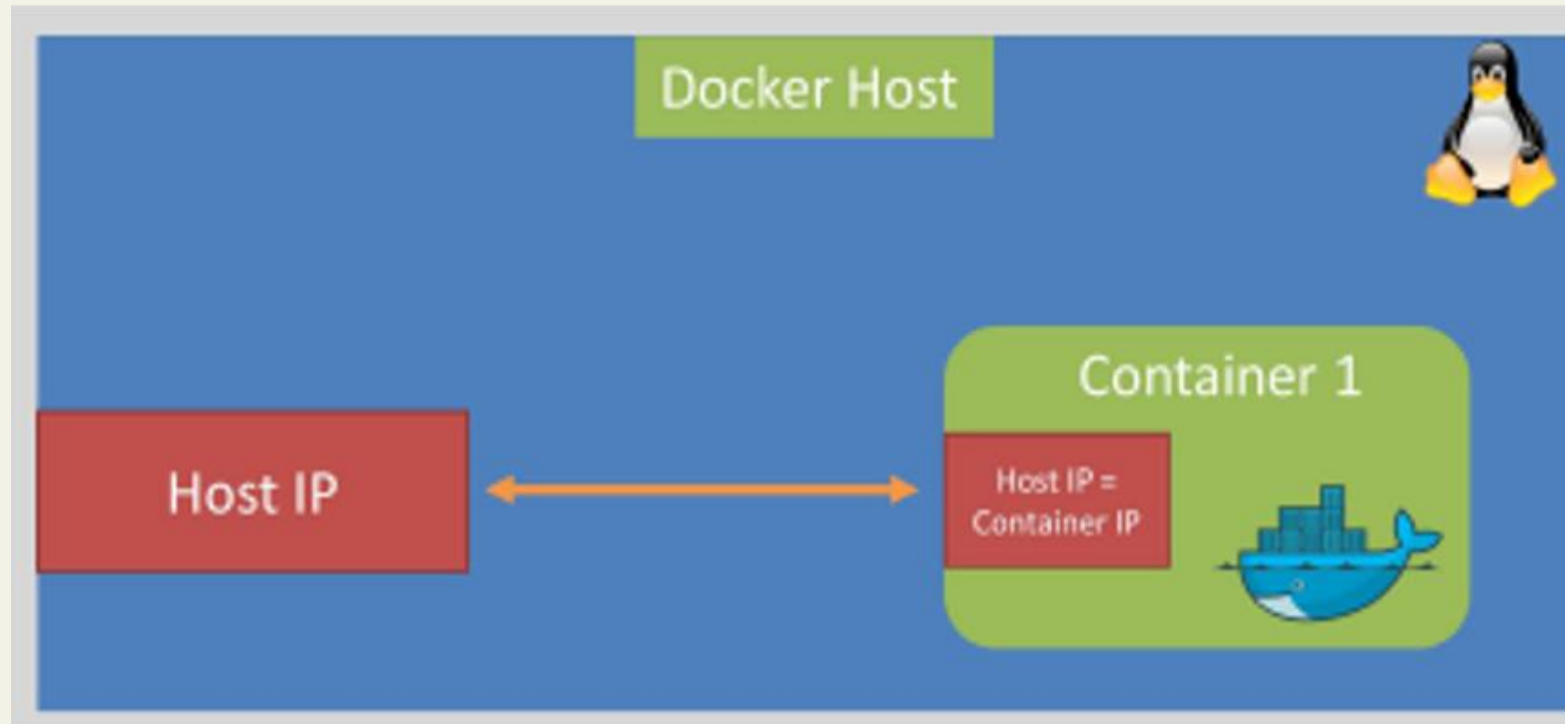


- ❖ How the network happens when we have multiple containers communicating to external networks through the host ?



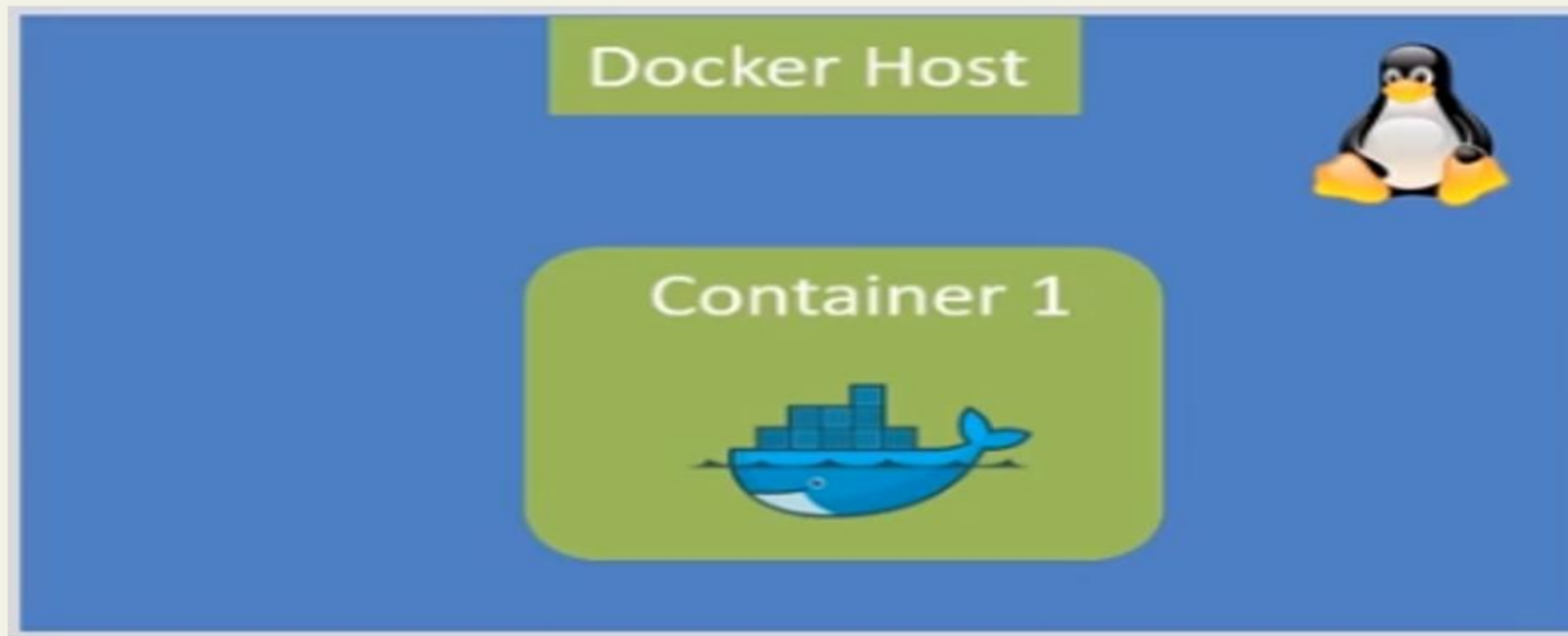
2-Host network mode :

- Disable docker's network isolation . It means containers see and use exactly the same network Interfaces the host has available.



3- none network mode :

- The none network driver does not attach containers to any network , containers don't access the external network or communicate with other containers.



Docker network commands:

```
docker network ls
```

```
docker network create <name>
```

```
docker network create - --subnet 10.0.0.0/16 <network-name>
```

```
docker network inspect <network --name>
```

Docker network commands:

`docker run -it --network <network-name> <image-name>`

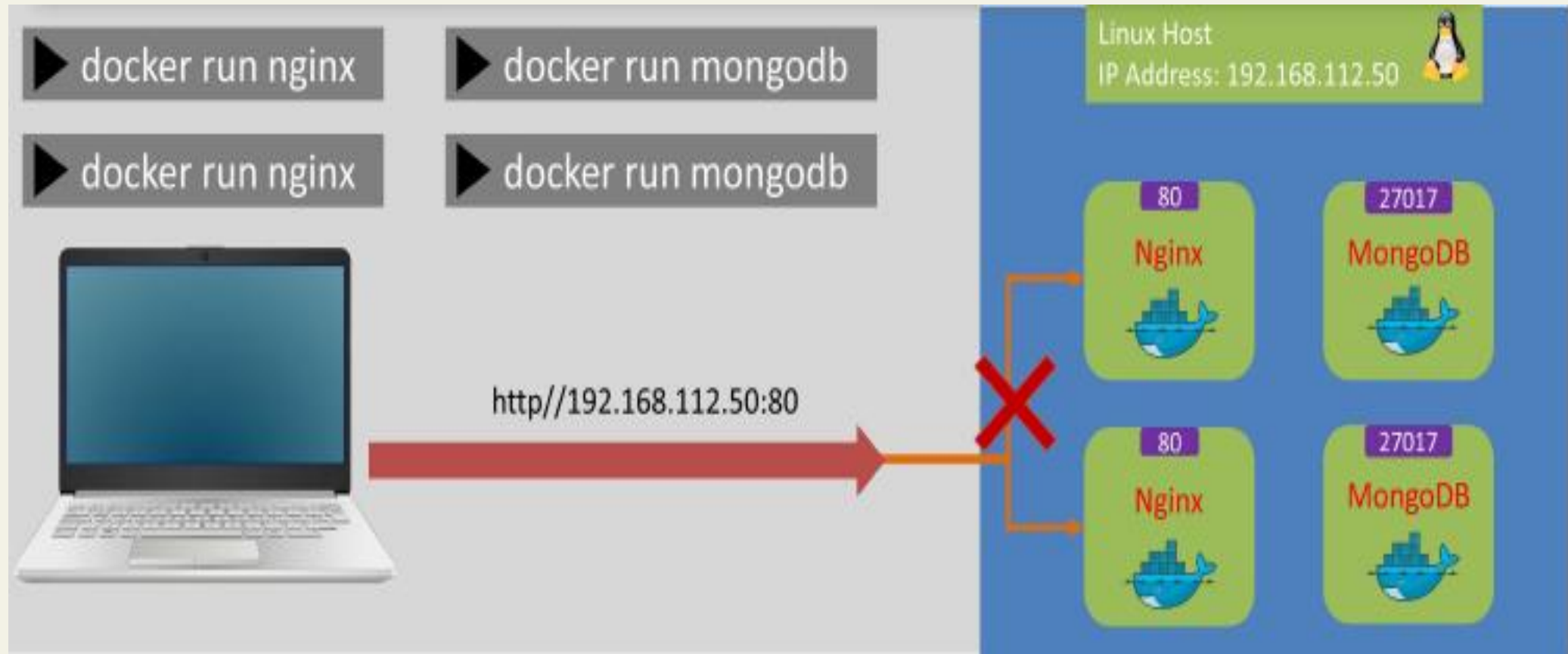
`docker network connect <network-name><container-name>`

`docker network disconnect <network-name><container-name>`

Lab4:

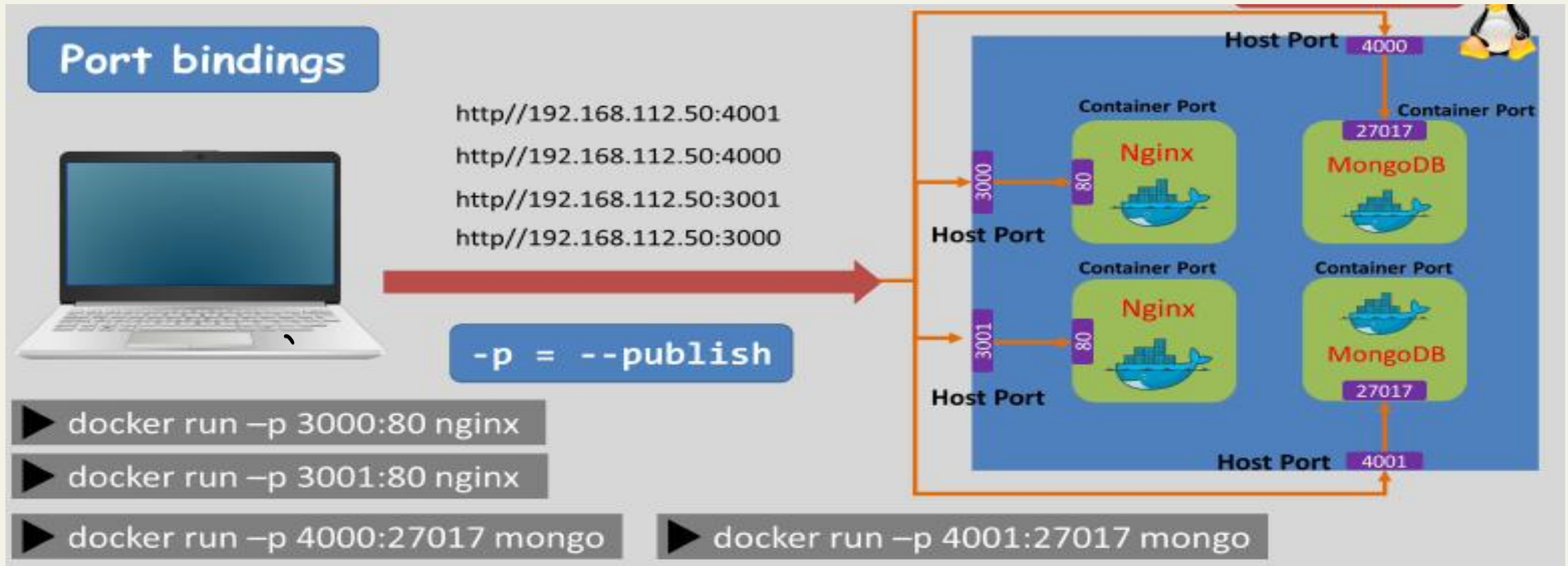


Container port:



Container port-port mapping:

```
docker run -p <host port>:<container port> <image-name>
```



Lab5:



Docker images :

```
FROM ubuntu  
RUN apt-get update  
CMD [ "echo" , "hello from container" ]
```

Docker image from build to runtime :

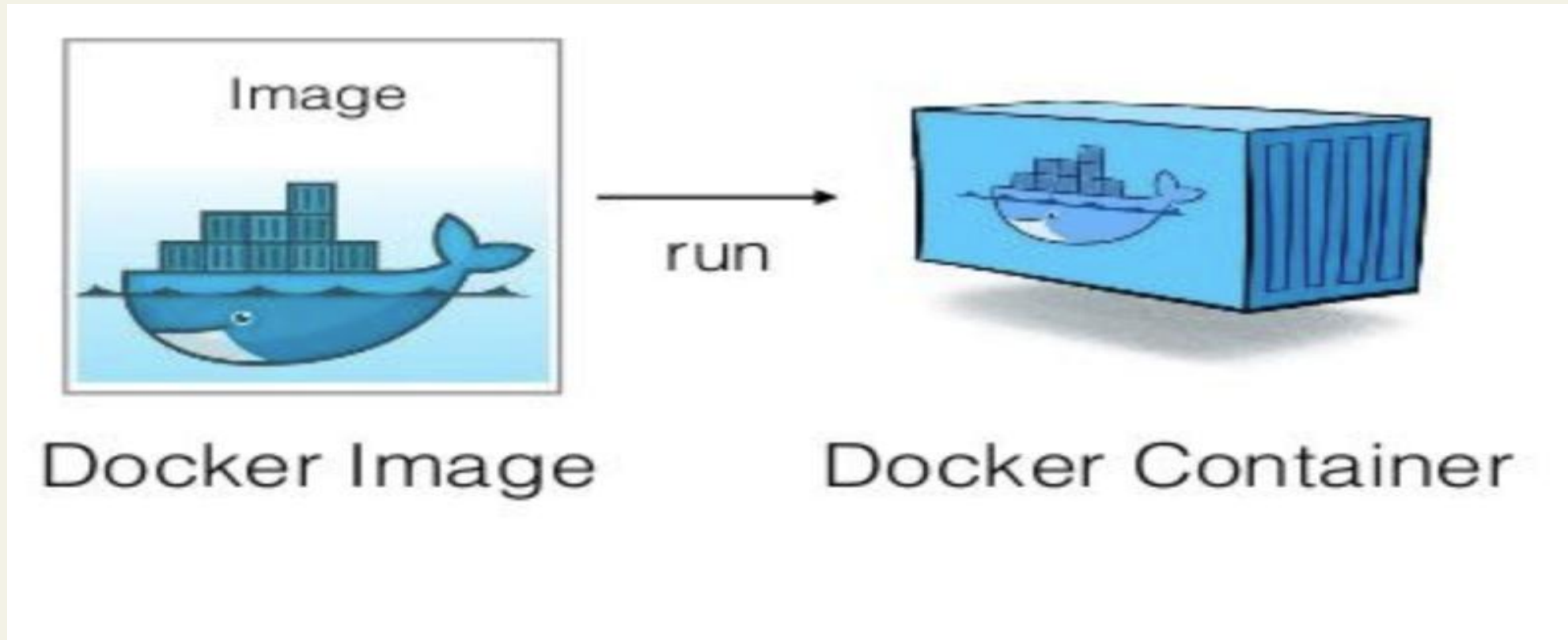
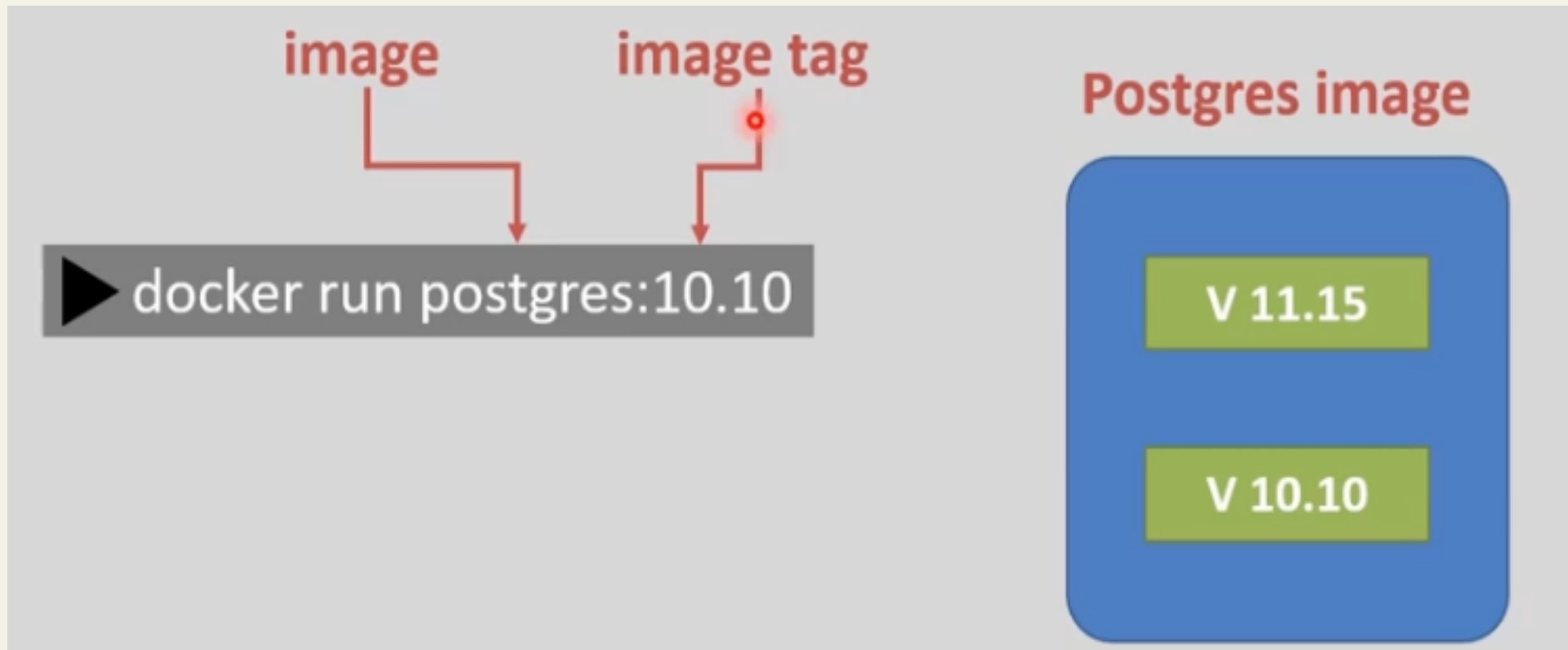


Image tag:



Lab6:



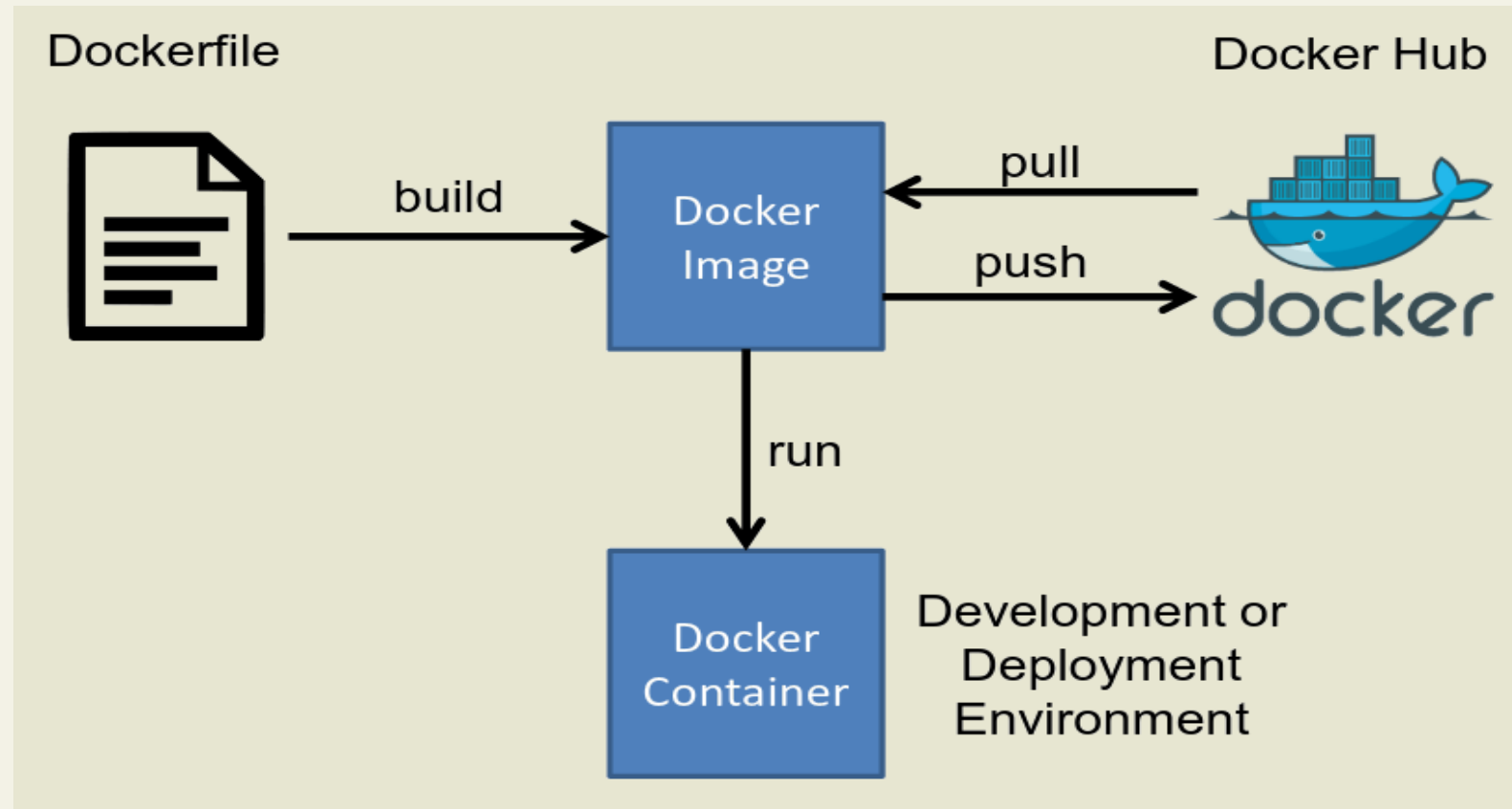
Docker images(cont.):

Docker run postgres:latest

Docker run postgres

docker history <image-name>

Docker registry:



Docker registry commands:

```
docker login
```

```
docker tag local-image:tagname [username-account]/repo-name:tagname
```

```
docker push [username-account]/new-repo:tagname
```

Lab7:

